

TECHNICAL ASSESSMENT

6–8

 HOURS (EST.)
7-DAY WINDOW

Conversational Threat Intelligence Analyst

AGENTIC AI

TAKE-HOME

MID LEVEL

Build an agentic AI application that a SOC analyst can query in natural language to perform threat analysis. This assessment evaluates three things together: agent engineering, security reasoning, and production judgment. You may use AI coding assistants — we expect it.

01 THE TASK

Build a chat-based threat intelligence agent. An analyst types questions in plain English; your agent interprets the intent, calls the appropriate threat-intelligence tools, correlates the results, and replies with evidence. Use free-tier APIs (e.g. VirusTotal, AbuseIPDB, AlienVault OTX, NVD, Shodan) or a mock dataset of your own.

IOC Lookup

Reputation of an IP, file hash, or domain — e.g. “Is 45.83.122.10 malicious?”

Actor & TTP

Profile a threat actor or its techniques — e.g. “What TTPs is APT29 known for?”

Exposure Reasoning

Map a software version to known threats — e.g. “We run Confluence 7.13 — are we exposed?”

Pivoting

Move between related entities — e.g. “Pivot from that IP to related domains.”

Multi-Turn Follow-Ups

Resolve references against prior context — e.g. “and what’s its ASN?”

02 CORE REQUIREMENTS

- A natural-language chat interface (CLI or web).
- Intent parsing that routes each query to the correct tool.
- Multi-turn context retention — resolves references like “it” or “that domain”.
- Evidence-grounded answers with source attribution — no fabricated intel.
- Prompt-injection resistance: both direct (typed in chat) and indirect (hidden inside retrieved data).
- Graceful handling of API errors and missing data.

- A clear README to install and run the application.

03 BONUS (OPTIONAL)

Eval / Test Harness

Show consistent behaviour across runs.

Confidence Scoring

Attach confidence to each finding.

Observability

Trace the agent's tool calls and steps.

Cost & Rate Limits

Handle quotas and control token spend.

04 VIDEO DEMONSTRATION — REQUIRED

Record a complete end-to-end demo of your working application.

Submit a single screen recording of the application running live. Demonstrate every core requirement and each query type working — with nothing failing. Briefly narrate what is happening as you go. A submission without a clean, complete demo video is considered incomplete.

05 WHAT TO SUBMIT

- A GitHub repository with a README.
- A short design note (max 1 page) covering intent routing and injection defense.
- The end-to-end demonstration video (file or link).

06 HOW WE EVALUATE

EVALUATION DIMENSION	WEIGHT
Intent parsing & tool routing	20%
Multi-turn context & state	15%
Threat-analysis soundness & grounding	20%
Security reasoning (injection + scope)	20%
Agent & tool engineering	15%
Conversational UX	10%